From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

(ii) **Section-B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. There is no negative marking.

(iii) **Section-C (01 – 06)** contains 6 Numerical based questions with single digit integer as answer, ranging from 0 to 9 and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

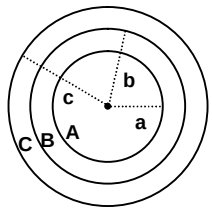
Physics

PART – I

SECTION – A
Single Correct Choice Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

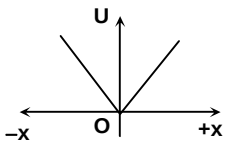
1. Three concentric charged metallic spherical shells A, B and C have radii a, b and c; charge densities σ, $-\sigma$ and σ; and potentials V_A, V_B and V_C respectively. Then which of the following relations is correct?

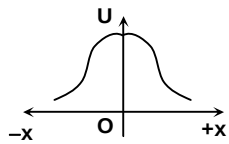


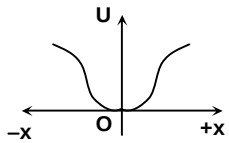
(A) $V_A = \dfrac{(a+b+c)\sigma}{\varepsilon_0}$ (B) $V_B = \dfrac{[(a^2/b)-b+c]\sigma}{\varepsilon_0}$

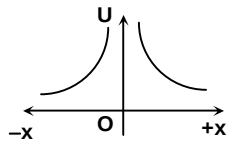
(C) $V_C = \dfrac{[(a^2+b^2)/c+c]\sigma}{\varepsilon_0}$ (D) $V_A = V_B = V_C = \dfrac{(a-b+c)\sigma}{\varepsilon_0}$

2. Four identical charges are placed at the four vertices of a square lying in YZ plane. A fifth charge is moved along X axis. The variation of potential energy (U) along X axis is correctly represented by

(A)


(B)


(C)


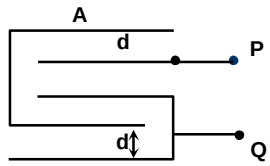
(D)


3. Two resistances 300 ohm and 400 ohm are connected in series with a 60 volt supply. A voltmeter connected across 400 ohm reads 30V. The same voltmeter when connected across the other resistance will read.

(A) 22.5 V (B) 50 V
(C) 30 V (D) 10 V

Space for rough work

4. The equivalent capacitance between P and Q in the figure is (area of each plate is A and separation between two consecutive plates is d)



(A) $\frac{A\varepsilon_0}{2d}$ (B) $\frac{3A\varepsilon_0}{2d}$

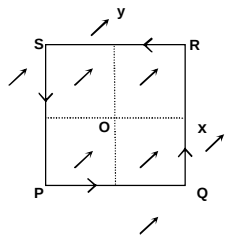
(C) $\frac{2A\varepsilon_0}{d}$ (D) $\frac{5}{3}\frac{A\varepsilon_0}{d}$

5. Two long parallel straight conductors carry currents i_1 and i_2 ($i_1 > i_2$). When the current are in the same direction, the magnetic field at a point midway between the wires is 20 μT. If the direction of i_2 is reversed, the field becomes 50 μT. the ratio of the currents i_1/i_2 is

(A) 5/2 (B) 7/3

(C) 4/3 (D) 5/3

6. A uniform constant magnetic field $\vec{B}$ is directed at an angle of 45° to the axis in the xy –plane. PQRS is a rigid, square wire frame carrying a steady current I_0, with its centre at the origin O. At time t = 0, the frame is at rest in the position as shown in figure, with its sides parallel to the x and y axes. Each side of the frame is of mass M and length L. The torque τ about O acting on the frame due to the magnetic field will be?



(A) $\frac{BI_0L^2}{\sqrt{2}}\left[+\hat{j}-\hat{i}\right]$ (B) $\frac{BI_0L^2}{\sqrt{2}}\left[-\hat{j}+\hat{i}\right]$

(C) $BI_0L^2\left[+\hat{j}-\hat{i}\right]$ (D) $BI_0L^2\left[-\hat{j}+\hat{i}\right]$

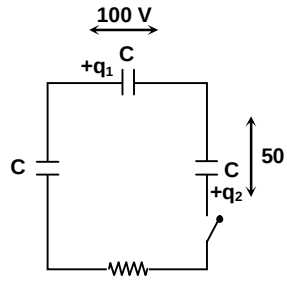
7. A coil is in the form of equilateral triangle of side 'a'. The current in the coil is I. The magnetic field at a distance 'a' equidistant' from all the vertices will be

(A) $\frac{\mu_0 I}{2\sqrt{3}\pi a}$ (B) $\frac{\sqrt{3}\mu_0 I}{2\pi a}$

(C) $\frac{\mu_0 I}{\sqrt{3}\pi a}$ (D) $\frac{\mu_0 I}{4\sqrt{3}\pi a}$

Space for rough work

8. Two capacitor each of capacitance C = 1 μF each charged to potential 100 V and 50 V respectively and connected across a uncharged capacitor 'C' = 1μF as shown in the figure if switch 'S' is closed then charge in the uncharged capacitor at equilibrium will be



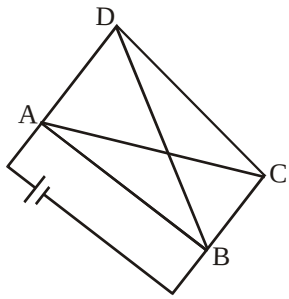
(A) $\frac{50}{3}\mu C$ (B) 50 μC

(C) 25 μC (D) $\frac{100}{3}\mu C$

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

9. A wire frame in the form of a tetrahedron ABCD is connected to a dc source. The resistance of all the edges of the tetrahedron are equal. Choose the correct statement (s)



(A) the electric current through DC is zero
(B) if we remove anyone of the edges AD, AC, DB or BC, the change in current remain same
(C) the current through AB is maximum
(D) the change in current will be same if AB or DC is removed

10. A parallel plate capacitor is charged to a certain voltage. Now, if the dielectric material (with dielectric constant k) is removed then.
(A) the capacitance increases by a factor k
(B) the electric field reduces by a factor k
(C) the voltage across the capacitor decreases by a factor k
(D) the charge stored in the capacitor increases by a factor k

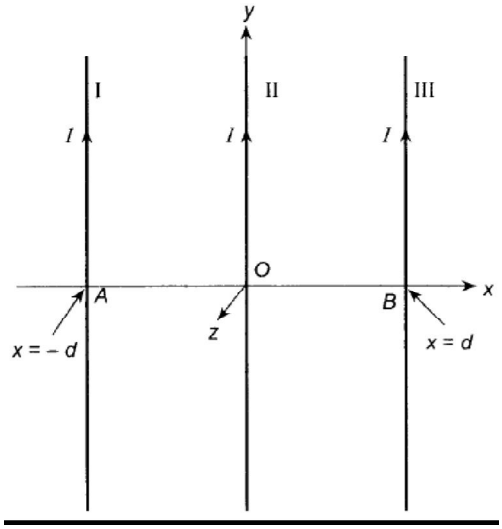
Space for rough work

11. Two point charges $q_1 = +2\mu C$ and $q_2 = -4\mu C$ are placed at A and B respectively as shown. The distance between q_1 and q_2 is 4cm



(A) A line of force emanates from q_1 making a angle 90° with A B enters q_2 at angle 60°.
(B) The net electric flux will be zero over any surface that encloses a volume excluding A & B
(C) The electrostatic potential is zero at infinitely many points in space
(D) None of these

12. Three infinitely long thin wires, each carrying current I in the same direction are in the x – y plane in a gravity free space (see Fig.). The central wire is at x = 0 while the other two wires are at $x = \pm d$. The net magnetic field
(A) cannot be zero beyond B (i.e. $x > d$)
(B) cannot be zero beyond A (I,e. $x < -d$)
(C) will be zero between x = 0 and x = d
(D) will be zero between x = 0 and x = – d

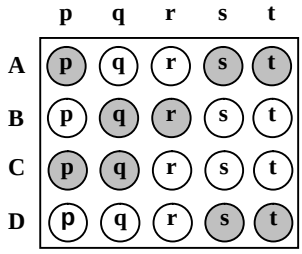


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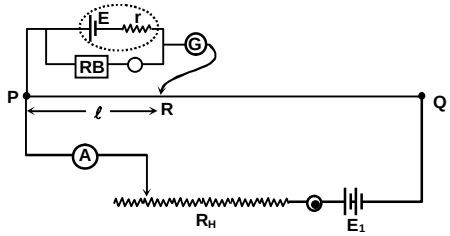
SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



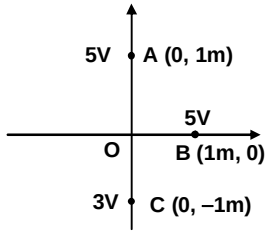
1. A potentiometer experiment setup is shown in the given diagram. P is the point when there is no deflection in the galvanometer. E_1 is the source. Match the following.



Column – I		Column – II	
(A)	If only emf of battery E_1 is increased	(p)	point P will shift to left
(B)	If only resistance of rheostat is increased	(q)	point P will shift to right
(C)	If resistance box is connected and increased.	(r)	point P will remain unchanged
(D)	If an identical battery is connected in parallel to E.	(s)	point P may shift to left or right

Space for rough work

2. An uniform electric field exists in straight line in the X–Y plane. The potential at different point in region are shown in the figure. Now match the following column if charge particles P (mass = 10^{-6} kg, charge $\sqrt{2}\ \mu C$) and θ (mass = 10^{-6} kg. Charge = $-2\sqrt{2}\ \mu C$ are released from origin)



	Column A		Column B
(A)	Co-ordinate of position of particle P at time t = $2\sqrt{2}$ sec	(p)	$(4\sqrt{2}$ m, $4\sqrt{2}$ m)
(B)	Co-ordinate of position of particle θat time t = 2 sec	(q)	$(-4\sqrt{2}$ m, $-4\sqrt{2}$m)
(C)	Distance travelled by particle Q in 2 sec is	(r)	8 m
(D)	Distance travelled by particle P in 2 sec is	(s)	4 m

SECTION –C
Integer Answer Type

This section contains **6 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. Three resistors of resistances 2 Ω, 3 Ω and 6 Ω are available to be connected across a battery of internal resistance 1Ω and emf. 4V by means of conducting wires. The maximum current (in amp) that can be drown through the cell by using all the given resistances is

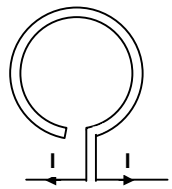
2. A capacitor of variable capacitance C = α t (where α is constant and 't' is time) is connected with a battery of emf E = βt. Rate of energy stored in capacitor is proportional to t^n. Then find the value of n. (t is time and α, β are constants)

Space for rough work

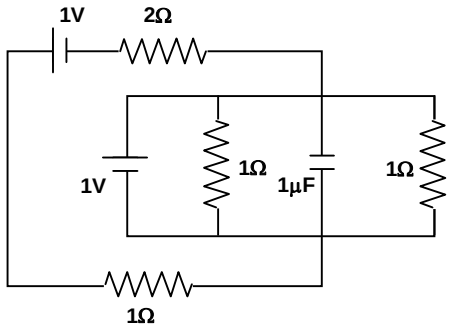
3. A wire of length L is wound on a spherical form, and a current I is passed through it. The coil is then placed in a uniform magnetic field of induction B. Find the number of turns so that the torque on it to be maximum.

4. A capacitor of variable capacitance $C = \alpha t$ (where α is constant and 't' is time) is connected with a battery of emf $E = \beta t$. Rate of energy stored in capacitor is proportional to t^n. Then find the value of n. (t is time and α, β are constants)

5. An electric current i is passed through a circular loop of folded copper wire as shown in the figure. Find the magnetic induction at the centre of the loop.



6. The charge on the capacitor in steady state is x μA. Calculate x.



Space for rough work

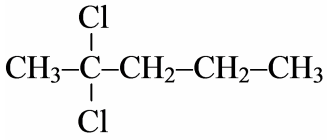
Chemistry

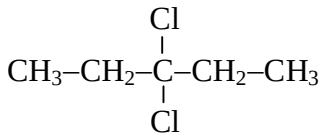
PART – II

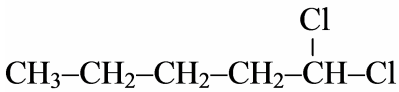
SECTION – A
Single Correct Choice Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A compound (A) $C_5H_{10}Cl_2$ on hydrolysis gives (B) $C_5H_{10}O$. The compound (B) reacts with NH_2OH, to form oxime and gives positive iodoform test but does not give Fehling's test. The compound (A) is

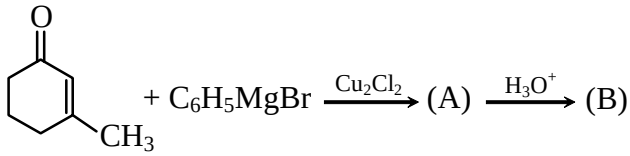
(A) 

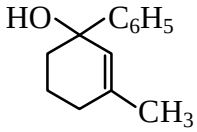
(B) 

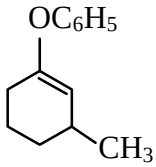
(C) 

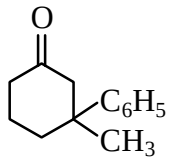
(D) None of these

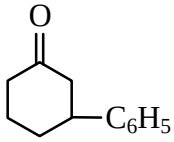
2. The major product (B) of the following reaction is



(A) 

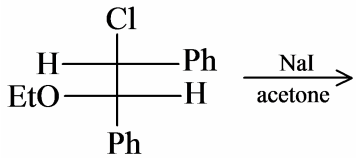
(B) 

(C) 

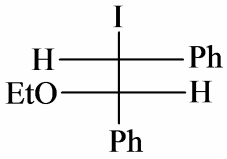
(D) 

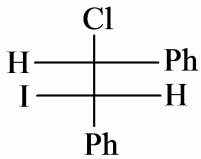
Space for rough work

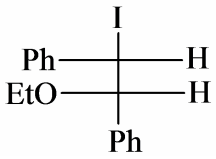
3. In the following reaction,

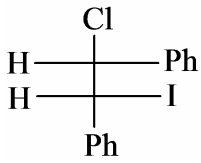


the structure of product is

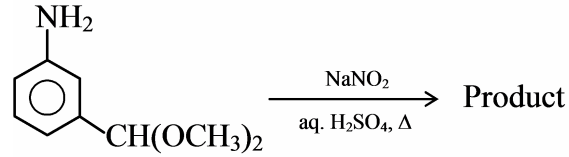
(A) 

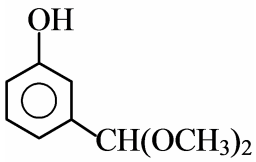
(B) 

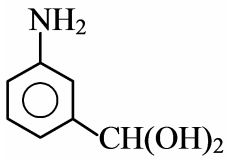
(C) 

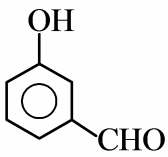
(D) 

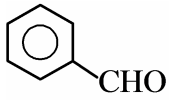
4. The major product of the following reaction is



(A) 

(B) 

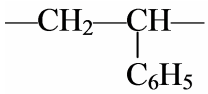
(C) 

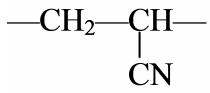
(D) 

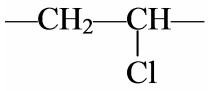
5. The presence of carbon in an organic compound is detected by heating it with
   (A) Sodium metal to convert it into NaCN.
   (B) CaO to convert it into CO which burns with a blue flame.
   (C) CuO to convert it into CO_2 which turns lime water milky.
   (D) Cu wire to give a bluish–green flame.

Space for rough work

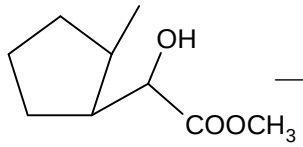
6. The repeat unit of polystyrene is

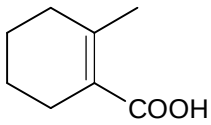
(A) 

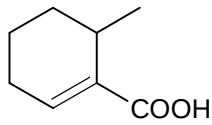
(B) 

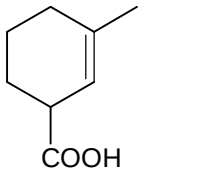
(C) 

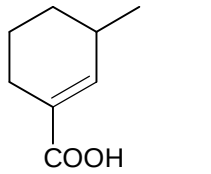
(D) $—CCl_2—CCl_2—$

7.  P; 'P' as a major product must be,

(A) 

(B) 

(C) 

(D) 

8. Benzene diazonium chloride on reaction with phenol in weakly basic medium gives …….. and the reaction is ……..
   (A) diphenyl ether, nucleophilic addition
   (B) p-hydroxy azobenzene, electrophilic substitution
   (C) chlorobenzene, electrophilic addition
   (D) benzene, elimination

Space for rough work

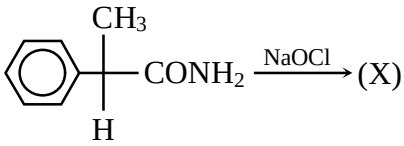
Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

9. Which of the following statement(s) about D(+) glucose is/are true?
(A) Naturally occurring glucose is dextrorotatory (B) It reduces ammoniacal $AgNO_3$ solution
(C) On polymerisation it forms galactose (D) It forms cyanohydrin on reaction with HCN

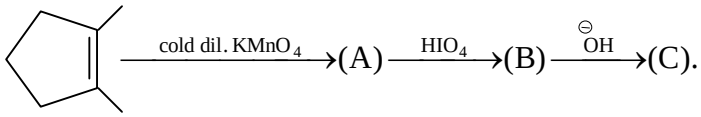
10. Which of the following statements is/are correct?
(A) In S_N2 mechanism, the rate determining step involves two species
(B) The S_N2 mechanism involves the formation of intermediate carbocation
(C) The S_N2 mechanism is accompanied with Walden inversion
(D) The S_N2 mechanisms may involve intramolecular rearrangement

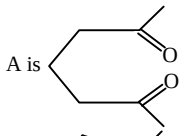
11. Which of the following statements are true about the major product (X) in the reaction?

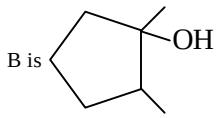


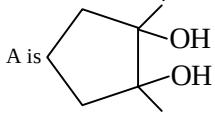
(A) (X) is a cyclic amide
(B) (X) has an asymmetric centre
(C) (X) is a primary amine
(D) (X) can also be obtained by treatment of NH_3 with $C_6H_5CH(CH_3)Br$

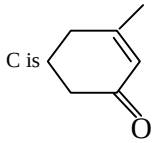
12. In a reaction sequence



(A) 

(B) 

(C) 

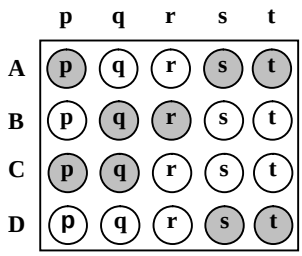
(D) 

Space for rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following:

Column - I (Substrate + reagent)		Column - II (Major product)	
(A)	$CH_3–CH_2–CH_2–Cl \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(p)	$CH_3–CH_2–CH=CH_2$
(B)	$CH_3–\underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{\displaystyle |}{\underset{\displaystyle |}{C}}}}–CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(q)	$CH_3–CH=CH–CH_3$
(C)	$CH_3–CH_2–\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}}–CH_3 \xrightarrow[BuOH]{t-BuO^-K^+}$	(r)	$CH_3–CH=CH_2$
(D)	$CH_3–CH_2–\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}}–CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(s)	$CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2$
		(t)	$CH_2=CH–CH=CH_2$

2. Match the following:

Column I		Column II	
(A)	Hell–Volhard-Zelinsky reaction	(p)	Preparation of amines
(B)	Reformatsky reaction	(q)	Preparation of β–hydroxy aldehyde
(C)	Claisen condensation reaction	(r)	Preparation of β–hydroxy ester
(D)	Schmidt reaction	(s)	α–halogenation of carboxylic acid
		(t)	Preparation of β–keto ester

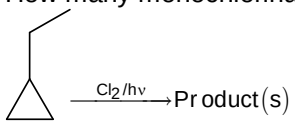
Space for rough work

SECTION –C

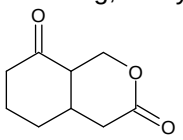
Integer Answer Type

This section contains **6 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

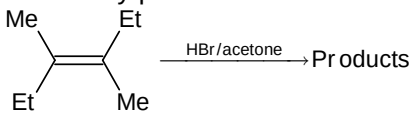
1. How many monochlorinated products (theoretically) of the following reaction is/are possible?



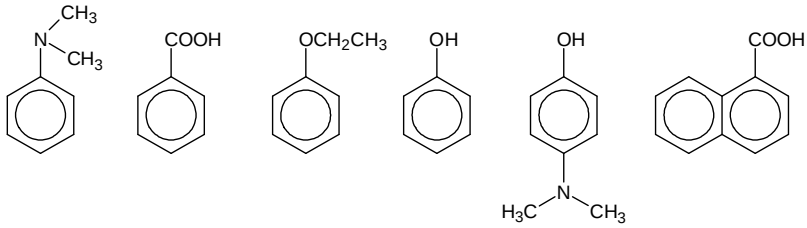
2. How many molecules of CH_3MgX are consumed in the reaction with one molecule of the following, in dry ether?

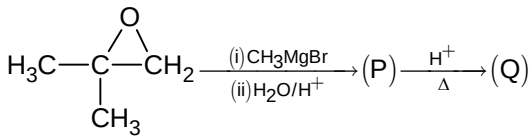


3. How many products will be formed in the following reaction?

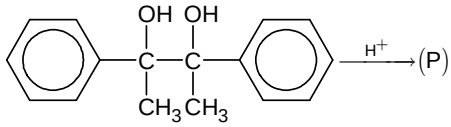


4. Amongst the following the total number of compounds soluble in aqueous NaOH is



5. 

How many hyperconjugating structures of "Q" are possible?

6. 

How many α - H to carbonyl groups are there in the product "P"?

Space for rough work

Mathematics

PART – III

SECTION – A
Single Correct Choice Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. If $f(x) = (a^2 - 3a + 2)\left(\cos^2\frac{x}{4} - \sin^2\frac{x}{4}\right) + (a - 1)x + \cos 1$ does not passes critical point then the possible value of a is/are
   (A) $a \in (0, 1)$ (B) $a \in (0, 2)$
   (C) $a \in (0, 4)$ (D) none of these

2. Let $f(x) = \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) - \ln(x^2 + x + 1) + (b^2 - 5b + 4)x + c$ be a decreasing function $\forall\ x \in R$, then the value of b is
   (A) $\left(-2, \frac{2}{3}\right]$ (B) (0, 3)
   (C) (3, 4) (D) (2, 3)

3. The area of the region described by a moving point P lying inside the circle $(x - a)^2 + y^2 = a^2$ above x-axis such that its distance from circle along the radius through P is greater than its distance from x-axis is
   (A) $\frac{4a^2}{3}$ (B) $2a^2$
   (C) a^2 (D) $\frac{2a^2}{3}$

4. If $y = \sqrt{a^2 - b^2\cos^2(\ln x)}$ then $x^2y^2\left(\frac{dy}{dx}\right)^2$ is equal to
   (A) $a^2 - y^2$ (B) $a^2 + b^2 - y^2$
   (C) $(a^2 - y^2)(b^2 - a^2 + y^2)$ (D) $(b^2 - y^2)(a^2 - b^2y^2)$

Rough work

5. $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+\sin^2 x}-\sqrt[4]{1-2\tan x}}{\sin x+\tan^2 x}$ equals to

(A) – 1 (B) 1

(C) $\dfrac{1}{2}$ (D) $-\dfrac{1}{2}$

6. If function f(x) satisfies $f'(x + k) + f'(x) = 0\ \forall\ x \in R$ and $k \in \left(0, \dfrac{\pi}{2}\right)$, then the value of $\dfrac{\int_a^{2nk+a}\left|\sin\left(\frac{\pi x}{2k}\right)\right| f'(x)dx}{\int_0^{2k}\cos\left(\frac{\pi x}{2k}\right) f(x)dx}$ is

(A) 0 (B) $n\pi$

(C) $-n\pi$ (D) $-\dfrac{n\pi}{2k}$

7. The domain of the function $f(x) = \cos^{-1}(\sec(\cos^{-1}x)) + \sin^{-1}(\text{cosec}(\sin^{-1}x))$ is

(A) $x \in R$ (B) $x = 1, -1$

(C) $-1 \le x \le 1$ (D) $x \in \phi$

8. Let f(x) be a polynomial one-one function such that $f(x) f(y) + 2 = f(x) + f(y) + f(xy)\ \forall\ x, y \in R - \{0\}$, $f(1) \ne 1$, $f'(1) = 3$. Let $g(x) = \dfrac{x}{4}(f(x)+3) - \int_0^x f(x)dx$, then

(A) g(x) = 0 has exactly one root for $x \in (0, 1)$ (B) g(x) = 0 has exactly two roots for $x \in (0, 1)$

(C) $g(x) \ne 0\ \forall\ x \in R - \{0\}$ (D) $g(x) = 0\ \forall\ x \in R - \{0\}$

Space for rough work

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

9. Let $f(x) = \dfrac{\left(x+\frac{1}{x}\right)^6 - \left(x^6+\frac{1}{x^6}\right) - 2}{\left(x+\frac{1}{x}\right)^3 + \left(x^3+\frac{1}{x^3}\right)}$ then

(A) range of f(x) is $R - (-6, 6)$ (B) f(x) is increasing $\forall\ x \in (1, \infty)$

(C) f(x) is decreasing $\forall\ x \in (-\infty, -1)$ (D) range of f(x) is $R - \{0\}$

10. Let f(x) and g(x) are two periodical functions then which of the following may be periodic

(A) f(x) g(x) (B) f(x) + g(x)

(C) f(x) – g(x) (D) f(g(x))

11. If $f(x) = \int_1^x \frac{\ln t}{1+t}dt$, then

(A) $f\left(\frac{1}{x}\right) = -\int_1^x \frac{\ln t}{t(1+t)}dt$ (B) $f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln t}{t(1+t)}dt$

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$ (D) $f(x) + f\left(\frac{1}{x}\right) = \frac{1}{2}(\ln x)^2$

12. The point of inflection for the curve $y = \dfrac{x+1}{x^2+1}$ is

(A) $\left(-2-\sqrt{3}, \frac{-\sqrt{3}-1}{4}\right)$ (B) $\left(-2+\sqrt{3}, \frac{-1-\sqrt{3}}{4}\right)$

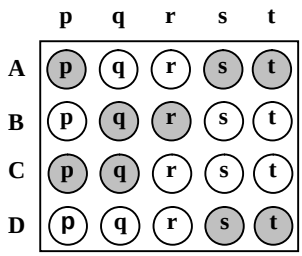
(C) $\left(-2+\sqrt{3}, \frac{1+\sqrt{3}}{4}\right)$ (D) (1, 1)

Space for rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Matching Column I with Column II:

	Column – I		Column – II
(A)	$\lim_{x\to0}\left(\frac{ae^x-b}{x}\right)=2$, then a + b is equal to	(p)	–1
(B)	$\lim_{x\to0}(\cos 2kx)^{\cot^2(x/k)}=e^{-2}$, then k is	(q)	3
(C)	$\lim_{x\to0}(1+ax+bx^2)^{1/x}=e^2$, then b is	(r)	4
(D)	$\lim_{x\to0^-}[x]\cos(2\pi[x])$ is equal to (where [.] denotes greatest integer function)	(s)	1

2. Matching Column I with Column II:

	Column – I		Column – II
(A)	The area of the figure bounded by $y = x^2$ and $y = \sqrt{x}$ is	(p)	4/3
(B)	$\int_0^4\left\{\sqrt{x}\right\}dx$ has the value ({x} denotes fractional part of x}	(q)	5/3
(C)	The area of the region for which $0 < y < 3 - 2x - x^2$ and $x > 0$ is	(r)	7/3
(D)	$\int_{-\pi/2}^{\pi/2}\sqrt{\cos x-\cos^3 x}\,dx$ equals	(s)	1/3

Space for rough work

SECTION –C

Integer Answer Type

This section contains **6 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. The number of solutions of $x + \int_0^x \ln t\,dt = \frac{x^2}{3}$, $(x \in R^+)$ is _____

2. If $g : D \to R$ be a function such that $g(x) = \underbrace{\ln \ln \ln \ln .....\ln}_{\text{(n times)}}(4(x^2 + x + 1) + \sin(\pi x))$, $(n \in N)$, then the least value of n for which g becomes onto, is _____

3. The number of points of discontinuity of f (x) in [0, 2] where
$f(x) = \begin{cases} [\cos \pi x], & x \le 1 \\ |2x - 3|\,[x - 2], & x > 1 \end{cases}$, ([.] denotes the greatest integer function), are _____

4. The area bounded by curves $y = \left[6 + x\left[\frac{1}{x}\right]\right]$, $y^2 - 18x + 18 = 0$ and $6x - 5y - 6 = 0$, (where [.] denotes the greatest integer function) is A, then $\frac{2A}{11}$ _____

5. The number of integral values of a ; $a \in (6, 100)$ for which the equation $[\tan x]^2 + \tan x - a = 0$ has real roots; where [.] denotes greatest integer function is _____

6. The value of $\frac{8\sqrt{2}}{\pi}\int_0^1 \left(\frac{1 - x^2}{1 + x^2}\right)\frac{dx}{\sqrt{1 + x^4}}$ is _____

Space for rough work